HelioCast TR

AI-Powered Solar Energy Forecasting for Turkey

Samsung Innovation Campus AI
Capstone Project Proposal

Group 2



Why HelioCast TR?

The Problem

Solar power generation is highly weather-dependent.

Forecast uncertainty increases at 24–72h horizons.

This creates challenges for grid balancing and energy market planning in Turkey.

Our Goal

Develop a reproducible ML pipeline for 24–72h hourly forecasts + 80% intervals.

Combine historical weather forecasts with licensed solar generation data.

Benchmark results against persistence baselines without data leakage.

Forecast Horizon: 24h · 48h · 72h

Use Case: Grid integration & decision support

Data Sources & Preparation

01 **EPİAŞ**

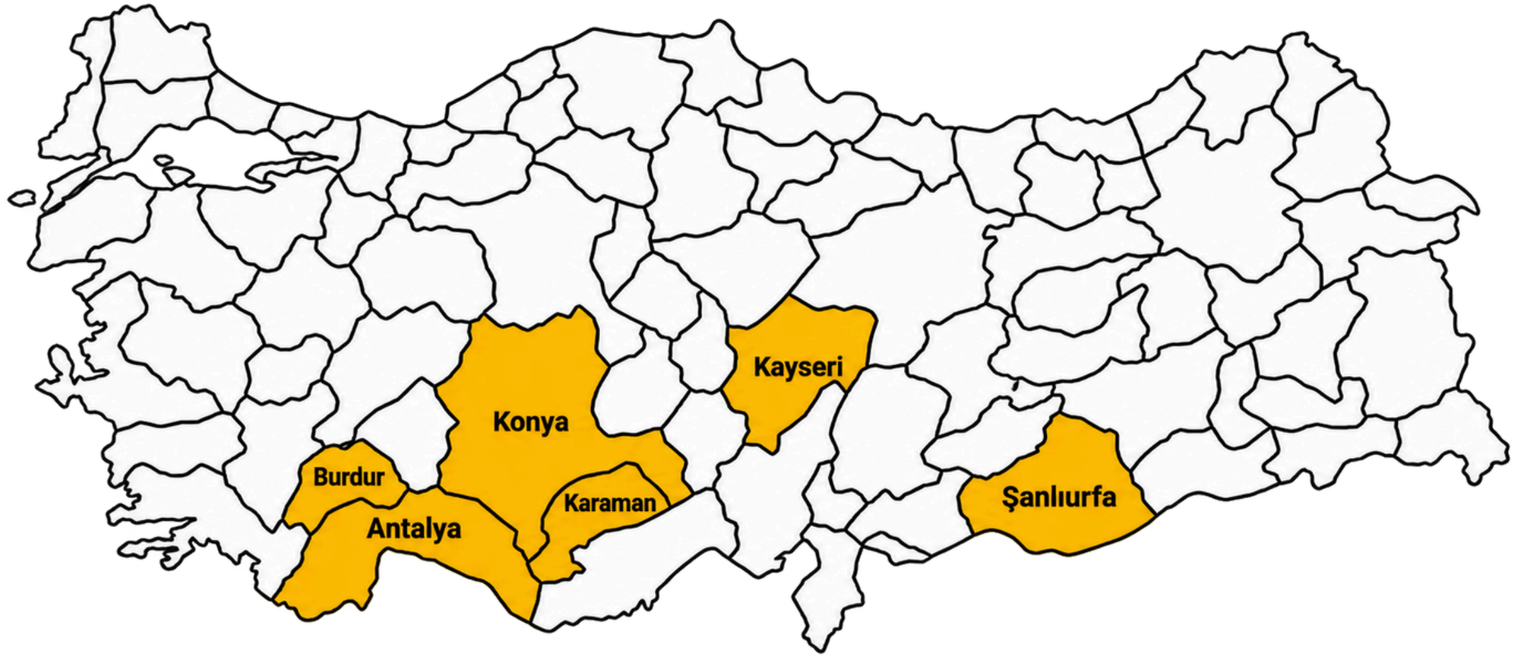
Hourly licensed solar generation, 2024–2026, normalized by capacity.

02 **Open-Meteo**

1–3 day-ahead forecasts: irradiance, cloud cover, temperature, humidity.

03 **Temporal Features**

Hour, day of year, and solar angles.



Chronological Splitting · Forecast-Time Alignment · No Data Leakage

How HelioCast TR Works

01

Data Inputs

Historical weather forecasts + solar generation

02

Feature Engineering

Weather, temporal, and solar features

03

ML Models

LightGBM / XGBoost

04

Forecast Output

24–72h Hourly Forecasts with 80% Intervals

Baselines

Persistence & Climatology

Explainability

SHAP

Validation

Chronological Train / Validation / Test Split

Evaluation & Sustainability

Model Evaluation

nMAE
Normalized Mean Absolute Error

RMSE
Root Mean Squared Error

Skill Score
Relative Error Reduction vs. Persistence

Evaluation Horizons: 24h | 48h | 72h

Daylight-only evaluation

Comparison against persistence and climatology

Sustainability Impact



SDG 7 — Affordable and Clean Energy
Supporting renewable energy integration and grid planning.



SDG 13 — Climate Action
Potentially reducing reliance on fossil-fuel backup generation.

From Prediction to Decision Support

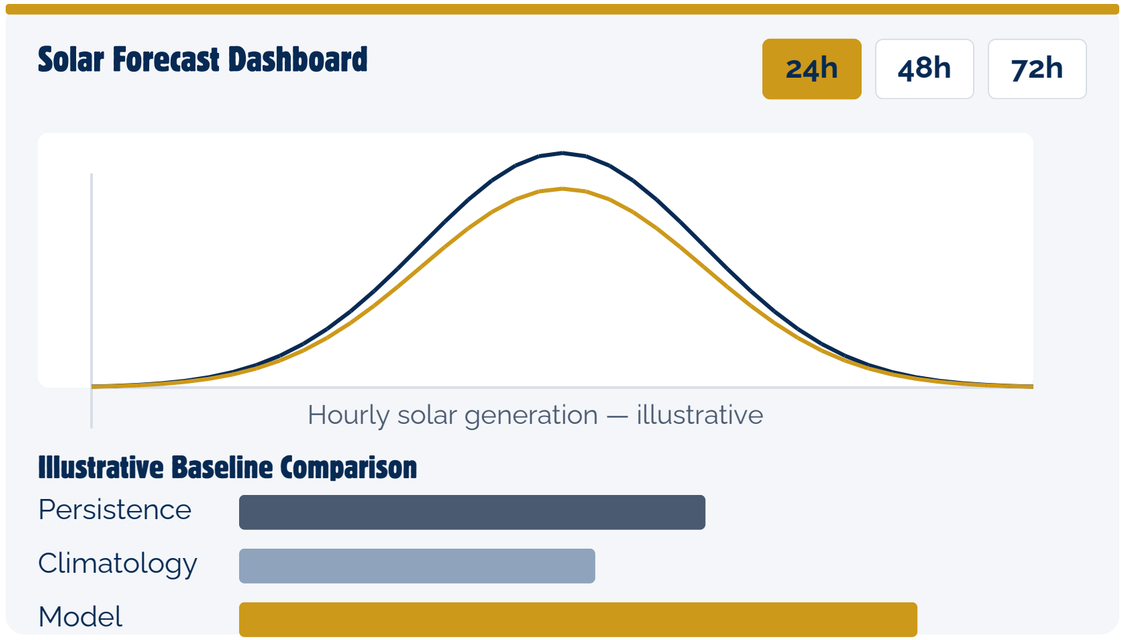
Expected Deliverable

Interactive Solar Forecasting Dashboard

- 24–72h forecasts with 80% intervals
- Actual vs. predicted generation
- Persistence baseline comparisons
- Interactive Plotly visualizations

Python · LightGBM / XGBoost · SHAP
Streamlit · Hugging Face Spaces

Dashboard Preview



Illustrative Layout — Results Pending · Bars: skill score

Ertekin (2020), Solar power prediction with an hour-based ensemble machine learning method.
Biçer (2026), Hybrid solar energy forecasting system integrating weather models and quantile regression.

Smarter Solar Forecasts. More Reliable Energy Decisions.